Sequences, Series and the Binomial Theorem — Index, Ratio and Convergence Errors

Answer Key

Algebra 2

Quick Answers

1. 51
2. 2, 2
3. 255
4. $x^4 + 8x^3 + 24x^2 + 32x + 16$
5. 24
6. 22
7. 270
8. —

Curriculum

Level: Algebra 2
HSA-APR.C.5 – *problems 4, 7*
HSF-BF.A.2, HSF-LE.A.2 – *problems 1, 2, 3, 5, 6, 8*
Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

1. *If they got 55: multiplied the common difference by n instead of by n minus one, so the count started at the wrong term*
2. *If they got 5: subtracted consecutive terms, which finds a common difference and not a common ratio*
3. *If they got 280: took the last term as a plus n times d rather than a plus n minus one times d, so the sum used a tenth term that is one step too far along*
5. *If they got 8: divided the first term by r instead of by one minus r*
7. *If they got 90: matched the index to the exponent on x and used two in place of three, so the power of the constant was taken from the wrong slot*

1. Find and fix: Dev's 12th term, $a = 7$, $d = 4$.

Step 1: name the slip. The first term is already in hand before any step is taken, so reaching the 12th term takes 11 steps, not 12. Dev multiplied d by n instead of by $n - 1$.
Step 2: use the explicit formula as written: $a_n = a + (n-1)d = 7 + (12-1)(4) = 7 + 44$.
Step 3: the 12th term is $\boxed{51}$.
Step 4: check by the smallest case the formula must survive — at $n = 1$ it gives $7 + 0 \cdot 4 = 7$, the first term ✓. Dev's version gives $7 + 4 = 11$ at $n = 1$, which is already the *second* term. That one-line test catches this error every time.
Watch for: 55, exactly one common difference too large.

2. Find and fix: Kai's ratio for 5, 10, 20, ...

(a) Step 1: a common *ratio* is what you multiply by, so it is found by division, not subtraction. Kai computed $10 - 5$, which is a common *difference* — the arithmetic tool applied to a geometric sequence.

Step 2: divide the second term by the first: $r = \dfrac{10}{5} = \boxed{2}$.

(b) Step 3: confirm on the next pair: $\dfrac{20}{10} = \boxed{2}$ ✓. The two ratios agree, so the sequence really is geometric with $r = 2$.

Step 4: why one pair is never enough — any two numbers have both a difference and a ratio, so a single pair is consistent with an arithmetic rule *and* a geometric one. Only the second pair separates them. (Kai's $r = 5$ would predict a third term of $5 \cdot 5 \cdot 5 = 125$, not 20.)

3. Find and fix: Ines's sum of ten terms, $a_1 = 3$, $d = 5$.

Step 1: her formula $S_n = \dfrac{n}{2}(a_1 + a_n)$ is correct. The error is in a_n: she used $a_1 + nd$, which is the *eleventh* term, so her series ran one term too far along.

Step 2: the true tenth term is $a_{10} = 3 + (10-1)(5) = 3 + 45 = 48$.

Step 3: $S_{10} = \dfrac{10}{2}(3 + 48) = 5 \times 51 = \boxed{255}$.

Step 4: check by the other form, $S_n = \dfrac{n}{2}\big(2a_1 + (n-1)d\big) = 5(6 + 45) = 255$ ✓.

Watch for: 280. The gap of 25 is exactly five times the common difference — the signature of an index that is one too high inside a sum.

4. Expand $(x+2)^4$.

Step 1: the binomial theorem gives the coefficients from row 4 of Pascal's triangle, $1, 4, 6, 4, 1$, with x losing a power and 2 gaining one at each step.

Step 2: term by term: $1 \cdot x^4$, $4 \cdot x^3 \cdot 2 = 8x^3$, $6 \cdot x^2 \cdot 2^2 = 24x^2$, $4 \cdot x \cdot 2^3 = 32x$, $1 \cdot 2^4 = 16$.

Step 3:

$$\boxed{(x+2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16}$$

Step 4: index check — the exponents on x and on 2 in each term are $(4,0), (3,1), (2,2), (1,3), (0,4)$: every pair sums to 4 ✓. A quick numerical check at $x = 1$: the expansion gives $1 + 8 + 24 + 32 + 16 = 81$, and $(1+2)^4 = 81$ ✓.

5. Find and fix: Nia's infinite series, $a_1 = 6$, $r = \frac{3}{4}$.

(a) Step 1: test convergence before using any formula. $\left|\frac{3}{4}\right| < 1$, so the terms shrink toward zero and the series converges — the formula is legal here.

(b) Step 2: name the slip. The denominator is $1 - r$, not r. Nia divided by the ratio itself.

Step 3: $S = \dfrac{a_1}{1-r} = \dfrac{6}{1 - \frac{3}{4}} = \dfrac{6}{\frac{1}{4}} = 6 \times 4 = \boxed{24}$.

Step 4: sanity check — the first few partial sums are $6, 10.5, 13.875, \ldots$, climbing steadily toward 24 and never reaching it. Nia's 8 is smaller than the third partial sum, so it cannot be the total.

Watch for: 8, i.e. $6 \div \frac{3}{4}$. Whenever $r < 1$, dividing by r rather than by $1 - r$ gives an answer far too small.

6. Find and fix: which term of $7, 11, 15, \ldots$ equals 91?

Step 1: the missing adjustment is the -1 in the index. The nth term is $7 + (n-1)(4)$, so the equation to solve is $7 + 4(n-1) = 91$, not $7 + 4n = 91$.
Step 2: solve it: $4(n-1) = 84$, so $n - 1 = 21$.
Step 3: $n = \boxed{22}$ — the 22nd term equals 91.
Step 4: verify directly: $7 + 21 \times 4 = 7 + 84 = 91$ ✓.
Teaching point: the uncorrected equation returns $n = 21$, a whole number, so nothing looks wrong. An index error hides best exactly when it produces a plausible integer — always substitute back.

7. Find and fix: the coefficient of x^2 in $(x+3)^5$.

(a) Step 1: in $\binom{n}{k} x^{n-k} p^k$ the index k counts the *constant's* power, not x's. The two exponents must add to 5, so x^2 forces $5 - k = 2$, giving $k = 3$. Vic set k to the exponent he could see.
Step 2: the term is $\binom{5}{3} x^2 3^3 = 10 \cdot x^2 \cdot 27$.
(b) Step 3: the coefficient is $10 \times 27 = \boxed{270}$.
Step 4: check the index rule on the term found: exponents 2 on x and 3 on the constant sum to 5 ✓.
Watch for: 90, from $\binom{5}{2} 3^2$. Note that $\binom{5}{2}$ and $\binom{5}{3}$ are both 10, so the binomial coefficient does not reveal the error — only the power of 3 does. That symmetry is why this mistake survives a re-check.

8. Explain: convergence, an impossible sum, and two habits.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.
(a) The partial sum of a geometric series is $S_n = a_1 \dfrac{1 - r^n}{1 - r}$. When $|r| < 1$, $r^n \to 0$ as n grows, so S_n settles on $\dfrac{a_1}{1-r}$ — that limit is what "the sum" means. When $|r| \geq 1$, r^n does not shrink: the terms stay the same size or grow, the partial sums increase without bound (or oscillate), and no limit exists. The formula is the limit of the partial sums, so quoting it outside $|r| < 1$ quotes a limit that is not there.
(b) Adding positive numbers can only increase a running total, so no partial sum, and hence no limit of partial sums, can be negative. A negative output is therefore a proof that the formula was applied where it does not apply: with $r = 2$, the denominator $1 - r$ is negative, which manufactures a sign the series itself can never produce. The condition $|r| < 1$ is not a formality — it is what makes the answer mean anything.
(c) Two habits: (i) *index* — test the formula at $n = 1$; it must return the first term, which instantly exposes nd written for $(n-1)d$; (ii) *ratio* — always divide at least two consecutive pairs and check they agree, since one pair is consistent with both an arithmetic and a geometric rule.
What a full-credit answer must contain: $r^n \to 0$ (or partial sums growing without bound) as the reason for the condition, an argument that a sum of positive terms cannot be negative, and two habits that are *checks* rather than restatements of the formulas.